

**AN INTELLIGENT YOUTUBE CONTENT FILTERING AND PERSONALISED
RECOMMENDATION SYSTEM FOR CHILDREN USING AGE-AWARE VIDEO
ANALYSIS**

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BONAFIDE CERTIFICATE

Certified that this project report titled **An Intelligent Youtube Content Filtering And Personalised Recommendation System For Children Using Age-aware Video Analysis** is the bonafide work of **SOUNDARYA K (2024178044)** who carried out the project work under my supervision. Certified further that to the best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents an intelligent video recommendation system designed specifically for children, focusing on safety, age-appropriateness, and content relevance. With the rapid increase in digital media consumption among young users, ensuring that children are exposed only to suitable and educational content has become a critical challenge. This system aims to address that issue using machine learning and content analysis techniques.

The proposed system integrates multiple modules including video ingestion, feature extraction, content classification, and recommendation generation. Videos are analyzed based on key attributes such as age suitability, topic category, difficulty level, and safety score. A supervised machine learning model is employed to learn from labeled data and predict these attributes for new, unseen videos. Additionally, speech-to-text techniques are used to extract and analyze video transcripts, enabling deeper semantic understanding of the content.

To generate personalized recommendations, the system uses vectorization and similarity-based approaches such as cosine similarity, ensuring that recommended videos align with the child's profile and previously consumed content. The architecture also incorporates a pre-analysis and caching mechanism to optimize performance and deliver real-time recommendations efficiently.

By combining machine learning, natural language processing, and intelligent ranking strategies, this project provides a scalable and effective solution for creating a safe digital environment for children. The system not only improves content filtering but also enhances engagement by recommending relevant and age-appropriate videos.

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